complyance

Take-home assignment: CRM and Zoho Books integration

Full-Stack Developer Intern, Complyance

1. The task

Sales works in a CRM. Finance works in Zoho Books. Build the integration between them, so that a won deal becomes a sales order, the sales order is invoiced, and the payments received show up back on the deal.

Create your own example deal for a made-up customer company of your choice, worth **AED 120,000 plus 5% VAT** and billed in three milestones: **50% on signing, 30% at go-live and 20% ninety days after go-live**.

The integration works in both directions:

- **CRM to Zoho Books:** a Closed Won deal creates the customer, the sales order and its milestone invoices in Zoho Books.
- **Zoho Books to CRM:** every deal in the CRM shows the status of its sales order, invoices and payments from Zoho Books.

2. Setup

- Use any real CRM on a free plan or free trial. Do not build your own CRM.
- Use a Zoho Books trial organisation.
- Write the integration yourself. Zapier, Make, n8n, Zoho Flow and built-in CRM–Zoho connectors are not allowed for the integration.
- Use made-up data only: fake companies, people and tax numbers.
- Keep secrets out of the repository.

3. Must-haves

1. **Sales order to invoices, end to end.** One deal goes from Closed Won to a sales order in Zoho Books, at least two invoices are raised against that sales order, and at least one payment is recorded. The result is visible in both systems.
2. **Traceability.** Every Zoho Books record links back to its CRM deal, and the deal links to its Zoho Books records.
3. **Zoho Books authentication** handled correctly, including token expiry, with the Zoho region and organisation set in configuration.
4. **Zoho Books API limits** respected, and rate-limit errors handled.
5. **Idempotency.** Running any step twice, or receiving the same event twice, never creates a duplicate in either system.

6. **An integration log.** Every call to Zoho Books and to the CRM is recorded and viewable on a page, with the deal it belongs to.
7. **Explanation.** You must be able to explain how you integrated the two systems and how sales orders become invoices in Zoho Books. Include in the repository:
   - an architecture diagram;
   - a sequence diagram of one deal from Closed Won to paid;
   - a field-mapping table: CRM field, Zoho Books field, which system owns it, and the direction it flows;
   - your answers to the questions in section 5.
8. **Automated tests** for the logic that handles money and matching.

4. Requirements

From the CRM to Zoho Books

1. Detect a deal entering Closed Won. Closing the deal in the CRM must never wait on Zoho Books.
2. Find the customer in Zoho Books, or create it, without ever creating a duplicate. When more than one customer could match, a person decides.
3. Map the CRM deal's products to Zoho Books items. Never guess an item.
4. Create the sales order, linked to the deal, with the correct currency and tax.
5. Raise the three milestone invoices against the sales order when each milestone is reached. The invoice amounts must add up exactly to the sales order total.
6. Every record is created exactly once, even if the process crashes part-way.
7. Retry temporary errors; stop on permanent ones. Failures are visible on the deal, with a way to retry after a fix.

From Zoho Books back to the CRM

8. Reflect changes in Zoho Books (invoices sent, payments recorded, invoices voided or deleted) on the CRM deal promptly, and also catch anything that was missed.
9. For each deal, show: sales order total, percentage invoiced, percentage collected, amount invoiced but unpaid, amount not yet invoiced, next due date and whether anything is overdue.
10. Write only the CRM fields the integration owns, and only when their value has changed.
11. Build a collections view: per customer, the outstanding total, aging buckets (current, 1–30, 31–60, 61–90, 90+ days), the oldest unpaid invoice and the deals it belongs to.
12. Have at least 15 customers and 20 deals running through the integration, covering fully and partly invoiced sales orders, partial payments, overdue invoices and a voided invoice.
13. Report how many Zoho Books API calls one deal uses from Closed Won to paid, and how many a sync uses when nothing has changed.

5. Questions to answer

- Zoho Books is unreachable for six hours on the last day of the month. What happens?

complyance

- The deal is reopened after the first invoice was sent. What should happen?
- The deal amount changes after the sales order exists but before the second invoice.
- The CRM sends the same Closed Won event three times, out of order with an earlier stage change.
- An invoice is voided in Zoho Books and re-issued under a new number.
- One payment is recorded against two invoices that belong to two different deals.
- You reach Zoho Books' daily API limit in the afternoon. What does Finance see?

6. What to submit

1. **Deployed link** to your working integration, including the integration log page. If it needs a login, include reviewer credentials.
2. **Public GitHub repository link**, with a README explaining how to set it up and run it, the CRM you chose, and the diagrams, mapping table and answers listed in Must-have 7.
3. **Demo video link** with a detailed explanation: one deal end to end in both the CRM and Zoho Books, the integration log, your architecture, and how sales orders become invoices in Zoho Books.

Keep your CRM and Zoho Books accounts active until we have reviewed your submission. For any questions, reply to the email this assignment came with.